Malthus and the Less Developed World: The Pivotal Role of India
Author(s): John C. Caldwell
Source: *Population and Development Review*, Vol. 24, No. 4 (Dec., 1998), pp. 675-696
Published by: Population Council
Stable URL: https://www.jstor.org/stable/2808021

Malthus and the Less Developed World: The Pivotal Role of India

John C. Caldwell

Malthus's ideas have survived the two centuries since his *First Essay* was anonymously published, and they remain an intellectual force to be reckoned with.[1] The purpose of this essay is to identify why this has been the case and to explore the unique role of the English-speaking world and of the connection between Britain and India in determining the survival of the principle of population. First it is necessary to examine the kind of society that produced the theory and those of its elements that survived time and transplanting.

Thomas Robert Malthus (1766–1834) was a man of his time but also a man who helped shape his time. John Maynard Keynes (1951: 101) placed him with John Locke, David Hume, Adam Smith, and William Paley as his predecessors, Jeremy Bentham as his contemporary, and John Stuart Mill and Charles Darwin as his successors "profoundly in the English tradition of humane science, . . . a tradition marked by a love of truth and a most noble lucidity, by a prosaic sanity free from sentiment or metaphysic, and by an immense disinterestedness and public spirit." It was essentially a secular tradition, in spite of the fact that Paley and Malthus were Anglican clergymen. It was also one in tune with the outlook and interests of the rising commercial middle class. Malthus himself, like most of his contemporaries seeking general social laws, would have added Isaac Newton to this list.

Malthus was also in tune with his predecessors in another way that he readily conceded. His argument that population was limited by the availability of food was commonplace in the eighteenth century (Winch 1992: xi), and so he has been attacked as a plagiarist as if this were his sole contribution (e.g. Fryer 1965: 70). In fact, his views on the relationship between food and sustenance were expressed much more strikingly than those of other commentators, as well as in a far more significant context.

Malthus lived in England at a time when the older mercantilist economic and demographic policies of protecting domestic production and augmenting population growth were being successfully attacked. One reason was the growth of Britain as a trading power. Another was the related growth in power of the middle class, to be given political form by the Great Reform Act of 1832. Mercantilism was essentially the program of autocratic, centralized states, and lingered longer on the continent. Adam Smith (1723–90) had enunciated a new economics, suited to emerging capitalist Britain, in 1776 in his *Wealth of Nations*. But his work lacked the major population component that was subsequently produced by Malthus. The two doctrines complemented each other and provided a "liberal" political agenda. Malthus wrote the *First Essay* during a decade when population probably grew at twice the rate that had been the case during the decade when the *Wealth of Nations* was written (12 percent decadal growth compared with 6 percent, according to G. Talbot Griffith's estimates; see Mitchell and Deane 1962: 5). England had a Poor Law, codified in Elizabethan times, whereby magistrates spent parish rates meeting the needs of destitutes. Uniquely, this allowed a monetary estimate to be put on the cost of providing for the very poor and the additional cost imposed by their children. But it also became a greater political issue as the eighteenth century drew to a close because of inflation associated with the French Wars.

Malthus also wrote at a time when parents in at least some Western countries were probably beginning to feel that large families represented a greater burden than before, and when some deliberately used contraception to reduce ultimate family size. Abbé J. A. Dubois, who left France in 1792 to spend over 30 years in South India, described, in contrast to India, pre-Revolutionary France: "Other nations [those in Western Europe] . . . regard the fruitfulness of their women with aversion. They are, moreover, not ashamed to resort to wicked and disgusting means of reducing or destroying it altogether . . . that they might not deprive themselves of the means of satisfying their ambition or of procuring the luxuries of life . . . " (Dubois 1906: 592–593). It is unlikely that the British of the 1790s were any less apprehensive than the French of fruitfulness, but they, like Malthus, were ashamed of resorting to birth control and, in contrast to the French, it was to be another three-quarters of a century before their fertility fell.

The importance of Malthus was that he perceived a self-regulating system explaining population fluctuations. This was a system that Charles Darwin was to generalize through all species. Human fecundity was so great that fertility could easily outstrip the growth in food production. This would lead to starvation and a rise in mortality, bringing population growth back to the level of the food supply. Such crises were usually averted because, in "civilized" regions like Europe, a preventive check came into operation: marriage was either postponed or averted, thus reducing women's level of childbearing. There were other positive checks, especially outside Christian Europe: warfare, epidemic disease, the frequenting of prostitutes leading to sterilizing disease, polygamy, and such vices as infanticide, abortion, and contraception. If one check declined, others would inevitably become more forceful. The model was more sophisticated than this simple description suggests, and it was the additional complexities that carried a political message. Although Malthus believed that the total societal level of mortality had not changed for centuries, he held that, in a class society, the lowest classes, nearer the breadline, were characterized by the highest mortality and were likely to die first in a subsistence crisis. Furthermore, the most ignorant were the most likely to spurn the preventive check and were likely to marry as soon as they had the possibility of providing the ensuing children with the barest sustenance. This meant that financial transfers under the Poor Laws allowed them access to more food and encouraged them to marry earlier and breed more rapidly. The additional children put greater strains on the food supply and inflated its price, thus restoring population equilibrium by making it harder for other sections of society, especially the social class next above them, to buy food, and so raising their mortality. The intended benevolent functioning of the Poor Laws thus did nothing to reduce mortality or misery but rather spread these conditions to worthier parts of society. To force such profligate couples to take responsibility, their abandoned illegitimate children should be allowed to die (Malthus 1992: 263). This interpretation ultimately led to the Poor Law Amendment Act of 1834 with its establishment of urban workhouses to replace rural outdoor relief. Curiously, it was to be the first of Malthus's theses to have an impact on the administration of India.

Malthus considered the only satisfactory escape from the population–subsistence trap to be the restraint of marriage accompanied by sexual continence before and outside marriage. He began by regarding such restraint as typifying the behavior of civilized societies, primarily Christian Europe, especially northern Europe. Increasingly he regarded it as a criterion of civilization and even as a major element in the civilizing process. He did, however, believe that passion between the sexes, marriage, and childbearing were of great importance, and that it was unfortunate that natural law meant they had to be constrained.

Nevertheless, such conflicts were part of a greater equilibrium, or, better, near-equilibrium model, that had both beauty and a deeper meaning. Adam Smith had promoted "the thesis that man, in following the prompting of his nature, unconsciously gives substantial expression to some parts of the . . . Plan" (Campbell and Skinner 1976: 4). This was certainly natural law, but Smith never makes it clear whether it was also Divine Law. There is no such ambiguity in the case of Malthus. He, like many of his Cambridge generation, was greatly influenced by William Paley, the theologian and philosopher (1743–1805, author of *Principles of Moral and Political Philosophy*, 1785), and both believed in God's design of a world tending toward equilibrium (see Winch 1992: xix–xx). The historian Henry Buckle (author of *History of Civilization in England*, 1857) said that Adam Smith's model would still remain valid without the statistics and illustrations (Campbell and Skinner 1976: 56), and Malthus clearly regarded the principle of population in the same light.

Malthus's belief that the principle of population was part of the divine plan is the key to our understanding of the aspect of his work that later generations found so incomprehensible: his opposition to contraception. He certainly regarded contraception as vile, insulting to womanhood, and un-Christian—the consensus view in his day, shared by all but a few peripheral dissidents. The preventive check needed and taught self-control. It was a civilizing instrument. It was necessary to overcome the indolence of the poor and the uncivilized. God's plans to elevate his people could be thwarted by devices that kept starvation at bay while leaving indolence to reign. The long road to civilization might not be begun (see Breton 1983: 249–250). There was a strong case against "immediate gratification," marriage could be looked upon as a prize for industry and virtue, chastity had a solid moral foundation, and Christians should avoid temptation (Malthus 1992: 218–221). Malthus's refusal to link his identification of the threat of population growth with an encouragement of contraception greatly increased the possibility that his views would win widespread public acceptance.

The acceptance of Malthus's doctrines

The impact of Malthus on the world depended at first on how he was accepted by his British contemporaries. A generation after the publication of the *First Essay*, the principle of population was little debated. The Irish famine of 1846–47 was rarely treated as a test of Malthus's ideas. His first biographer, Bonar (1924: 363), looking back from 1885, believed that this was because most influential people, and indeed a broad swathe of society, were almost immediately converted to the central Malthusian thesis—a view subsequently taken also by Keynes (1951: 100–101), Glass (1953: 30), and Hollingsworth (1983: 213–215). England was a cost-accounting "nation of shopkeepers," predominantly Protestant and increasingly scientific and secular in much of its thought, growing crowded, with most families fearing too many births. Bonar reported that Malthus soon won over leading Whig politicians (William Pitt, the younger, Henry Brougham, James Mackintosh, and Samuel Whitbread), theologians (William Paley and Edward Copleston), economists (David Ricardo and Nassau Senior), and historians (Henry Hallam). In opposition were the poets, writers, and romantics (Robert Southey, William Hazlitt, and William Cobbett). Many supporters of the working class, however, remained hostile specifically because of the argument that acceptance of the principle of population led to hostility toward poor relief, and generally because of the judgmental attitude toward the value of the various social classes. A survey of nineteenth-century English diaries and novels showed that, throughout the century, "the British never saw very large families as particularly desirable. . . . [A]t least from the beginning of the nineteenth century, those who had substantially more children than was typical were seen as comic, inept, or sad victims of fate. . . . [T]he theme is one of reproduction outstripping resources" (Kane 1995: 152–153). Hollingsworth (1983: 217–218) believes that Malthus's thesis that disaster occurred when population growth outstripped food supply was so taken for granted by the 1840s that few thought of the Irish famine as testing Malthus. But it is likely that the acceptance of Malthusian concepts explained the delayed reaction of the British government to the famine.

Even after passage of the 1834 Poor Law Amendment Act, Malthus's influential successors spelled out his message about the dangers of helping the unfortunate in the clearest terms. In 1848 John Stuart Mill concluded in his *Principles of Political Economy:* "Everyone has a right to live. We will suppose this granted. But no one has a right to bring creatures into life, to be supported by other people" (Mill 1848: 252). Two decades later Walter Bagehot, the editor of *The Economist*, was applying "the principles of natural selection to political society":

> The most melancholy of human reflections, perhaps, is that, on the whole, it is a question whether the benevolence of mankind does most good or harm. Great good, no doubt, philanthropy does, but then it also does great evil. It augments so much vice, it multiplies so much suffering, it brings to life such great populations to suffer and to be vicious. (Bagehot 1869: 188–189)

John Carey (1992) has shown that by the last decades of the nineteenth century and the first decades of the present century the mainstream view of the English literary intelligentsia was strongly Malthusian. Population growth, especially of the poor, was dangerous.

Perhaps Malthus's most effective support came from where he least sought it, those who advocated artificial birth control. By 1822, with Malthus still alive and producing new editions, Francis Place in England had drawn the conclusion in his *Illustrations and Proofs of the Principle of Population* (1994) that the solution to the Malthusian difficulty was contraception, and he was probably the author of the "Diabolical Handbills," which circulated the following year with clear instructions to working-class women on how to practice contraception. By 1832 Charles Knowlton in the United States had published a book on the subject. The birth control movement saw where Malthus's views led, and it was not inappropriate that Charles Bradlaugh should found the Malthusian League in the early 1860s or that first the Dutch and then the English should describe contraception as neo-Malthusianism from the late 1870s on.

Malthus on the "less civilized" parts of the world

In the *First Essay* Malthus's treatment of the "less civilized" world, variously and unsatisfactorily now known as the third world or the less developed countries, was limited. Indeed, like most educated Englishmen of his time, he knew more about the Classical World. He drew on literature on both to draw contrasts with contemporary, Christian Europe. The latter was "civilized," not least, as he noted in the fourth edition, because "in modern Europe the positive checks to population growth prevail less, and the preventive checks more than in past times, and in the more uncivilized parts of the world" (quoted in Wrigley 1983: 115). In the uncivilized lands there was more warfare, epidemic disease, and vice, as well as famine. Indeed the first three more often than not kept starvation at bay. He identified un-Christian practices like infanticide, and even polygyny, as taking the place in Asia of Europe's restraint from marriage.

Malthus was well aware that in sparsely settled lands with few resources population was just as likely to press on resources as in the most densely settled countries. Indeed, he devoted more attention to North American Indians, Australian Aborigines, and Polynesians than to the heartland of agrarian Asia—partly, admittedly, because explorers and voyagers with a scientific bent were more frequently reporting on the former in the second half of the eighteenth century. But it was inevitable that later generations would most readily identify Malthusian crises as existing in densely populated agrarian societies, such as those of India and China, where famines and epidemics could kill millions.

Malthus wrote little about India, in spite of the fact that—or perhaps because—he was Professor of Political Economy, teaching India's future administrators, at the East India Company's College at Hertford and then Haileybury from 1805 until his death in 1834. He referred more frequently to East Asia. In the *First Essay* he pointed out that China was entirely occupied by tillage so that even the marginal lands, producing less, were being employed. He noted: "It is said, that early marriages very generally prevail through all the ranks of the Chinese. Yet Dr Adam Smith supposes that population in China is stationary" (Malthus 1986a: 25). If this were the case, then the explanation was famine and the "exposure" of children, for there did not appear to be a high level of disease. With regard to infanticide, he noted: "it is difficult to avoid remarking, that there cannot be a stronger proof of the distresses that have been felt by mankind for want of food, than the existence of a custom that thus violates the most natural principle of the human heart" (Malthus 1986a: 25). By 1826, in the sixth edition, he identified with alarm "the extraordinary encouragements that have been given [in China] to marriage" (Malthus 1986b: 128). He gathered evidence also of early and nearly universal marriage in India, as a result of which: "The population would thus be pressed hard against the limits of the means of subsistence, and the food of the country would be meted out to the major part of the people in the smallest shares that could support life. In such a state of things every failure in the crops from unfavourable seasons would be felt most severely; and India, as might be expected, has in all ages been subject to the most dreadful famines" (Malthus 1986b: 119–120).

From the second edition onward, Malthus became increasingly interested in the lack of prudence about marriage in the uncivilized nations, and in the conditions that nevertheless stopped them from being in a perpetual state of famine. Apart from disease, he found vice: fighting, unchastity and sterilizing disease, polygyny, infibulation, infanticide (see Godelier 1983: 134), as well as prolonged lactation and sexual abstinence within marriage. He saw the possibility that one day "the diffusion of education and knowledge" might bring into force the preventive check, the delay of marriage.

But those who administered the British colonies, especially India, drew three major lessons from Malthus. The first was that colonial administration reduced the level of the positive checks, certainly warfare and infanticide, and probably disease, with the result that populations would tend to grow faster, calling more frequently on famine as the ultimate check. The second was that interventions in cases of famines might, like poor relief in England, do more harm than good. The third was that the check of delayed marriage might not come into being for a very long time. In time, some were to argue that the Malthusian theses led directly to the need for birth control.

The Anglo–Indian link

The link between Britain and India, unique in colonial history, is of great importance for understanding the repercussions of Malthus and his doctrines (see Caldwell and Caldwell 1986: 37–43). The East India Company was chartered at the beginning of the seventeenth century and had established a presence in India within a decade. Two-and-a-half centuries later, in 1858, the British Government assumed direct control of the country, to relinquish it in 1947 to independent India and the newly created state of Pakistan. Even before 1858 the British Government had ultimate responsibility for India, and the affairs of the East India Company and the Government were intermeshed. India was not an ordinary colony; its population, standing at the middle of the nineteenth century at seven times greater than Britain's and constituting over 90 percent of the population of the colonial empire, ensured that. It had an ancient history recorded in literature and fascinated most Englishmen who had contact with it. And most of those who had been influential in the acceptance of the principle of population had such contact. Malthus worked nearly his whole adult life for the East India Company, as did John Stuart Mill for the India Office (where Keynes later worked briefly); Walter Bagehot's father-in-law was the Secretary for the Treasury in India. The administration of India, partly based on Moghul procedures, influenced the organization of Britain's own civil service.

Indian society was hierarchical, and upper-class Englishmen understood India's royal families and upper castes. They did not have to hurdle cultural and linguistic barriers in doing so, especially after the 1835 resolution within the framework of the Government of India Act with its educational provisions molded by Lord William Bentinck, the Governor-General, and Lord Thomas Macauley, which resolved to spend the Education Fund solely on English-medium education to promote European literature and science. Within a century 220,000 Indian schools and colleges were teaching 13 million students (Cunningham 1941: 149–150). Seeley (1885: 253) described the decision as "the great landmark in the history of the British Empire considered as an institution of civilization." English had been a medium of communication between the rulers and the ruled long before, but the 1835 resolution ensured that a huge English-speaking segment of the population would be conversant with British writings and thought, and would impinge on the evolution of Britain much as the British would leave their mark on the Indian intelligentsia (see O'Malley 1941; Rawlinson 1941). No other densely settled underdeveloped country had such a relationship with a European nation: the Dutch did not develop it with Java and neither China nor Japan was colonized.

The material circumstances of India had begun to make the British see it in a Malthusian way even before the publication of the *First Essay*: Sir Thomas Munro, in a 1795 letter to his sister, talked of famine and the check imposed by food production on population growth (Ambirajan 1976: 5). The British and Indians had common ground in Malthusianism. Both were densely settled. Once, densely settled lands had been regarded as evidencing prosperity, as was the case when the fourteenth-century Arab traveler, Ibn-Batuta, had reacted to the then most closely settled part of India by calling it "Golden Bengal." Malthus ensured that educated Englishmen and Indians would never again see the world this way.

Generations of British administrators and generations of reports ensured that India was known to Western researchers in a way that no other agrarian civilization was. There was no equivalent elsewhere in the underdeveloped world to the demographic statistical information that flowed from the Indian census beginning in the late 1860s. In India the Malthusian model could be tested at work in a non-European country as nowhere else.

India before the censuses

The first impact of Malthus's ideas on India's administrators came from his theses about the Poor Law, as has been well demonstrated by the historian of this phenomenon, S. Ambirajan (1976). A starkly clear parallel existed between the payment of Poor Law relief from Britain's upper classes to the destitute poor, and the payment of famine relief in India from revenues ultimately deriving from taxes contributed disproportionately by the better-off and upper castes to those at the bottom of the social ladder. Malthus's views and language were well known from early in the century, and in 1829 the Madras Board of Revenues of the East India Company commented on regional data appearing to show food production increasing faster than population: This is "at variance with one of the best established truths of political economy" (quoted in Ambirajan 1976: 6).

In the first two decades after Britain assumed direct responsibility for the administration of India, parts of the country were subject to devastating famine, particularly in Orissa in 1866 and Bengal in 1874. Ambirajan found that during these critical years many in the administration were converted to a Malthusian view of the situation. Lord Edward Lytton, Governor-General during 1876–80, was convinced that famine should be solved by the market, telling the Legislative Council in 1877 that the Indian population "has a tendency to increase more rapidly than the food it raises from the soil" (quoted in Ambirajan 1976: 6).

Many reports during the 1870s were written in Malthusian terms, arguing that famine was becoming commoner because the *Pax Britannica* had diminished many of the other direct checks, certainly warfare, infanticide, and suttee, and possibly epidemic disease. This view was expressed by the *Report Consequent to the Partial Census Conducted in India after the 1877 Famine* and by the 1880 Famine Commission. An 1881 report concluded that 80 percent of famine victims were drawn from the poorest 20 percent of the population, and if such deaths were prevented this stratum of the population would still be unable to adopt prudential restraint. Thus, if the government spent more of its revenue on famine relief, an even larger proportion of the population would become penurious. Ambirajan (1976: 7–11) concluded that, although the administration was divided about the value of famine relief, these Malthusian views undoubtedly affected its volume and timing. Although the language employed originated in Malthus's works, the authorities cited were more often his midcentury interpreters, especially Mill (1848) and Bagehot (1869). Sir Robert Giffen, statistician, Walter Bagehot's assistant editor of *The Economist*, and later head of the British Board of Trade, said he was expressing the view of most leading public men and economists when he wrote in the late nineteenth century that "'from the point of civilization and progress' millions of money spent in pauperizing Indians could not 'be contemplated with any satisfaction'" (quoted in Ambirajan 1976: 9–10).

British Empire censuses, like those of Britain itself, were planned to take place in the first year of each decade. In fact, India's first census, that of the 1870s, spanned the years from 1868 in the Punjab to 1876 in Bengal. But from 1881 on, censuses were held simultaneously in all parts of the country. The series of censuses, now spanning 120 years to 1991, are a demographic record for the less developed world without parallel. There was nothing similar for China until 1984 when the 1953, 1964, and 1982 censuses were released simultaneously. The Indian census reinforced interest in Malthusian equilibria, and later played a central role in creating the world's first national family planning program.

The reports of the censuses of the 1870s and 1881 were published separately by state or territory, usually under the auspices of a British census-taker in directly ruled British India and an Indian census-taker in the Indian States. Their reports were almost identical, those of persons well educated in the British tradition and understanding the population question as it was debated by British nineteenth-century political economists. From 1891 there was also an overall Indian report written by a British official, often recruited from Britain to take the census.

The census reports took as a major theme the race between population and India's ability to produce food. The 1868 Punjab report (Indian Census 1870: 17) warned that "as the limit of cultivation is approached, the rate of increase of population will halt." The report of Bengal (Indian Census 1878: 84) explored the central Malthusian issues: "British rule has established peace and security throughout the country, and so far has removed some of the causes which were at work to check the natural increase of the people. But Bengal is not more fertile than it used to be, and, if there has been an extension of cultivation, there are also large exports of agricultural products which were unknown at the beginning of the century." In less densely settled British Burma, it was largely the relaxation of the checks that was of interest: "These tracts of country came into our possession in the last war [the Third Burmese War], and in 1872 they had for 15 years been enjoying the benefits of a secure government and abundant material prosperity. One inevitable result of such a change is a rapid natural increase among the indigenous population" (Indian Census 1875: 19). In progressive Cochin, A. Sankariah noted, "The subjects of the Rajah of Cochin have therefore no need to fear of their numbers outstripping the capacity of the earth to sustain them if they will only exhibit ordinary prudence and enterprise" (Indian Census 1877: 56).

The outlook of the 1881 census reports was inevitably affected by the famines of the 1870s. The report for the Northwest Province and Oudh commented that the Benares Division "as a whole suffered less than the rest of the province in 1878–79 and consequently the positive checks to population have been less active" (Indian Census 1882: 26). The Baroda Territories report discussed "the laws of increase" and "the law of growth" and commented: "As a rule, in crowded cities and towns the preventive and also the positive checks operate in a greater degree than they do in the rural districts. Diseases and epidemics make their appearance more frequently in crowded cities and towns than they do in the rural districts. The preventive causes, namely moral restraint and vice, or rather compulsory restraint and vice, have a greater operation in the cities and towns than they have in the districts" (Indian Census 1883a: 73). The 1881 Madras report went beyond marriage to comment on famine and fertility: "It may be accepted that when food is scarce there are fewer births; whether this is exclusively the result of prudence, and whether that prudence is deliberate or instinctive, it is not here necessary to enquire" (Indian Census 1883b: 25–26). The Mysore report noted that famine mortality was particularly high among the Indian poor because even less help is transferred between castes than between classes (Indian Census 1884: 40).

In the all-India or General Report of the 1891 Census, J. A. Baines took a particular interest in "The Movement of Population" in Chapter 3. He believed that "population tends to equilibrium," and noted: "The conclusions arrived at by Malthus . . . have been severely criticised, and often misquoted or misapprehended, but, in the main, they have not been disproved" (Indian Census 1893: 58). This came from a man who had in his keeping the records of a census of over 200 million persons and two previous decennial censuses for comparison. When Baines wrote the report, Britain had been experiencing a contraception-induced fertility decline for twenty years. He knew of the decline but in common with most educated Englishmen of his time not its causes, and followed Herbert Spencer in believing that fecundity was being reduced by "the increased demand upon the nervous and intellectual qualities of life" (Indian Census 1893: 58). However, he followed Malthus in believing that a long period of enlightenment is needed "before the preventive checks can come into being." He noted that in India at that time, female marriage was early, but the cause of "the enormous number of births" was its universality (Indian Census 1893: 59–60). The 1911 Census Report tried to relate the cultivable area in different parts of India to the rate of population growth in order to explore Malthusian pressures (Caldwell and Caldwell 1986: 37).

The 1921 Superintendent of the Census, in an era when the impact of deliberate birth control in the West had become recognized, wrote in his report that India "has abandoned—or more or less abandoned—the old-fashioned methods of limiting population to an optimum, viz. periodic abstention from intercourse, abortion and infanticide and she has not yet adopted the methods of advanced countries, viz. postponement of marriage and voluntary birth control. She is at a point where the population is controlled by disease and disease only" (Indian Census 1924: 48–49). By the 1930s, J. H. Hutton, the Superintendent of the 1931 census and the author of a major study of the Indian caste system, wrote in his Report:

> It appears to be the general opinion of Indian economists who discuss the population problem of this country that the only practical method of limiting the population is by the introduction of artificial methods of birth control, though it is not easy to exaggerate the difficulties of introducing such methods in a country where the vast majority of the population regard the propagation of male offspring as a religious duty and the reproach of barrenness as a terrible punishment for crimes committed in a former incarnation. . . . Nevertheless, a definite movement towards artificial birth control appears to be taking place and is perhaps less hampered by misplaced prudery than in some countries which claim to be more civilized. (Indian Census 1933: 31–32)

He also echoed the later editions of Malthus in noting that improvements in child survival and education would assist fertility restraint.

The next full census was in 1951, in an independent but partitioned India, with an Indian head of the census, R. A. Gopalaswami. His attitudes were similar to those of his English predecessors, but sharpened by the fact that he was a citizen of this newly independent country, by his experience as Secretary of the Bengal Famine Inquiry, and by "the incredible population increase of the past 30 years." He spoke of "improvident maternity" and quantified the race between population and food (Huxley 1960: 72). He devoted his Chapter 4 to "Growth of Population: Checked and Unchecked" (Indian Census 1953). He drew attention to the inadequacy of "the natural checks on population growth, famine and disease" and to the fact that the principal and more satisfactory check in Europe was now contraception. It had been on the receipt of the preliminary results of the 1951 census together with Gopalaswami's interpretation of them that Prime Minister Nehru made his final decision to announce the world's first national family planning program.

Others also drew on the censuses to examine the Indian population situation. P. K. Wattal, of the Indian Audit and Account Service, published two books almost half a century apart, both concentrating on census data and reports, and both starting with Malthus and the Law of Population (Wattal 1916, 1958).

The impact on India of Malthusian and neo-Malthusian thought

The British and Indian elites were in intimate contact: both spoke English, they were educated from the same syllabuses, and their newspapers drew on the same news services. University education in the Presidency colleges, molded on English institutions, went back to the 1850s. The proportion of Indians in the higher levels of government, the Indian Civil Service (Indian Administrative Service, after Independence), increased steadily, but particularly rapidly after the 1935 Government of India Act. Interviews with retired members of the service suggest that by the 1930s few of either the English or Indian members doubted that the huge and growing Indian population threatened India's food supplies and progress. The emerging political elites shared this view: Jawaharlal Nehru had often been heard to say that "India would have been a much more advanced nation if its population were about half its actual size" (quoted in Mamoria 1961: 388). There was little in the strongly secular thought of either the British or Indian ruling classes or administrative elites that opposed Malthusian or neo-Malthusian solutions. Even Gandhi's reservations about the methods to be used echoed Malthus's feelings.

Documents that the Civil Service regarded as their own began to favor government intervention to slow population growth to an extent that was unparalleled anywhere else in the world. Hutton, writing the report of the 1931 Indian Census (1933), had raised the issue of government involvement in family planning. Shortly afterwards, Sir John Megaw, the former Director of Medical Services in India, wrote a report saying that Britain should introduce family planning into India (Sanger 1938: 464). Such views were supported in 1945 by the Final Report of the inquiry into the Bengal Famine (India Famine Inquiry Commission 1945: 385–386), and in 1946 by the prestigious Bhore Committee's investigation of India's health needs (India Health Survey and Development Committee 1946: vol. 2, pp. 483–487). These documents were all drawn upon for a 1947 report by the population subcommittee of the Congress Party's National Planning Committee, which was published as a book, and which advocated government intervention in favor of family planning in the medical and informational sectors (Shah 1947: 14, 113). All these documents drew on the Malthusian discussions in the censuses and other official documents, but, as in Britain in the mid-nineteenth century, the Malthusian conclusions had become so much a part of mainstream thought that there was often neither mention of, nor debate with, their progenitor. An American, Kingsley Davis (1946), wrote a paper that was to be influential in Indian planning circles and that posed the Malthusian dilemma of either "a calamitous rise in the death rate or a fall in fertility" (p. 243); he expected the former (p. 253) although he did identify some restriction of fertility arising from the Hindu taboo on widow remarriage (p. 254). These writings set the stage for Prime Minister Nehru's announcement in 1951 that India would establish a national family planning program from the beginning of the 1952 five-year planning period—an announcement met without objection except by the Gandhian Minister for Health, who objected to artificial means of fertility control.

Parallel developments occurred within influential nongovernmental sectors of the population. In 1916 a students' Eugenics Association was formed at Madras Presidency College. Two members of the association, the Maharajah of Mysore and Lady Rama Rao, thereby received their first exposure to neo-Malthusian ideas, which were to lead to the former establishing the world's first government birth control clinics in Mysore from 1930 and the latter becoming the founding president of the International Planned Parenthood Federation in 1953 (Caldwell and Caldwell 1986: 39). D. Ghosh (1946) published *Pressure of Population and Economic Efficiency in India*, replete with Malthusian concepts and references to "checks" and "moral restraints" (Ghosh 1946: 102 ff). Baljit Singh (1947) expressed the Malthusian and Gandhian view that the program for population growth constraint should center on "social attitudes towards marriage, sex relationship and family based on an elevation of social values " (Singh 1947: 128).

The 1935 annual meeting of the All-India Women's Conference called for a birth control program and invited Margaret Sanger to address their 1936 meeting "to assist them to put teeth into" their proposals (Sanger 1938: 461). India was the key to developing-world movements toward fertility control. Thus, although an unsuccessful meeting was held in 1948 in Cheltenham to attempt to set up an international family planning movement, a very successful meeting was held in Bombay (Mumbai) four years later. It gained great strength from its location in India, the strong delegation that Lady Rama Rao was able to bring from the All-India Women's Conference, the organizing involvement of Margaret Sanger, and above all, the fact that Nehru had already announced that India would have a national population program. He sent a sympathetic, but carefully worded message to the meeting. As a result of the 1952 conference the International Planned Parenthood Federation was established the following year; it had three developing-world members, India and two Asian British colonies, Singapore and Hong Kong.

Discussion

I want to elaborate upon two points addressed above: (1) Malthus and the Anglo–Indian connection and (2) the significance of Malthus in the move to curb developing-world population growth over the last half-century.

It is no accident that Malthus wrote in English. The way had been prepared by Adam Smith and others. But the way had been prepared for Smith by the growth of trade in the English-speaking world: in North America, throughout the British Empire, and in the North Atlantic Triangular Trade system.

It had also been prepared by the success of the trading middle classes in England in the 1642–45 civil war, the 1688 Glorious Revolution, and the 1832 Great Reform Act, and in the 1776–83 American Revolution and the 1789 Constitution. All these changes and Malthus's own writings occurred in a Protestant religious context, the latter specifically in an Anglican one. Without Thomas Robert Malthus, there would soon have been another, but he might not have written with such clarity and almost certainly without such a conjunction of ideas.

The English-speaking world became the nineteenth-century cradle for antinatalist thought. A partial exception to that generalization is the neo-Malthusian activities in the Netherlands, but they were backed by little indigenous political economic philosophy and the Dutch had no close interrelations with Indonesia of the type that developed between Britain and India. The main theater for playing out these antinatalist interests was Britain's huge colony, India. The establishment of the Indian national family planning program in 1952 was, in turn, of prime importance in the development of the world antinatal movement; its pattern of moral leadership by political and social elites and its top-down organization became a model for Asian programs (see Caldwell 1993). By this time, links in the fertility control field were being developed with India by experts from America as well as from Britain. Margaret Sanger had chosen India and Japan in the 1930s as the non-Western countries most likely to heed her message. When Nehru set up a committee in 1951 to advise on the need for, and design of, the family planning content in the 1952–56 Five Year Plan, they sought advice and reports from three Americans, Kingsley Davis, Pascal Whelpton, and William Ogburn (Caldwell and Caldwell 1986: 40). Davis had just published his *Population of India and Pakistan* (Davis 1951).

The impulse begun by Malthus continued and partly explains the central importance of English-language writers in stirring interest in world population growth after World War II and the early involvement of American and British technical aid in family planning. By 1965 the citizens of former British colonies in sub-Saharan Africa knew far more about family planning than did those in other countries in the region (Caldwell 1966); and indeed four of these countries, South Africa, Zimbabwe, Botswana, and Kenya, led the regional fertility decline, South Africa from 1960, the other three from the 1980s.

In spite of rapidly declining death rates worldwide after World War II and evidence that food production was increasing as fast as population, the most widely read literature with probably the greatest policy impact focused on the race between population and food supplies. Malthus's name was frequently employed.

William Vogt wrote in his *Road to Survival* (1949: 72): "And hand in hand with famine will stalk the shade of that clear-sighted English clergyman, Thomas Robert Malthus." Indeed, the clergyman stalks much of the book. Sir John Russell in *World Population and Food Supplies* (1954: 5) named, as thinkers in the field, Malthus and Sir William Crookes, who in an 1898 address pointed out that wheat eaters were increasing faster than wheat production. Russell's analysis did, however, present a fairly optimistic picture, at least in the short term. Karl Sax began *Standing Room Only* (1955) with the "Malthusian Laws."

By 1968 Paul Ehrlich opened *The Population Bomb* with the sentence, "The battle to feed all humanity is over." Interestingly, although he pursued Malthusian themes, even that of the class differential in starvation (p. 17), he, unlike most other popular authors on this theme, made no reference to the inventor of the principle of population. In contrast, Garrett Hardin, in "The tragedy of the commons," attributes much to Malthus's leadership. He exceeds the master in declaring that "Freedom to breed is intolerable" (Hardin 1968: 1246) and that the solution is "Mutual coercion mutually agreed upon" (p. 1247), but comes closer to Malthus's belief in self-restraint in the section of his essay on "How to legislate temperance?" (p. 1245).

More-recent writings are distinguished by an approach that is understandably more global and less national than Malthus's, but that adds to the problem of food those of water, erosion, air pollution, absorption of waste, and increasing scarcities in the forest and energy sectors. This was the approach of the Club of Rome (Meadows et al. 1972). It is also that of Lester Brown, as, for instance, in "The new world order" (Brown 1991), which does, however, give food supplies centrality with ample reference to Malthus (e.g., Brown 1991: 15 ff).

The divergence from Malthus has been greater in the academic literature, where most competing theory dates only from the 1940s. Most demographers accept some form of Frank Notestein's demographic transition theory, which placed a greater emphasis on the role of fertility in deciding whether societies survived or not. That theory was summarized by Ansley Coale and Edgar Hoover in the introduction to their book *Population Growth and Economic Development in Low-Income Countries* (1958: 9–10):

> [In an agrarian peasant economy] death rates usually fluctuate in consequence of variations in crops, the varying incidence of epidemics, etc. In such an economy birth rates are nearly stable at a high level. Death rates are high as a consequence of poor diets, primitive sanitation, and the absence of effective preventive and curative medical practices. High birth rates result from social beliefs and customs that necessarily grow up if a high death-rate community is to continue in existence. These beliefs and customs are reinforced by the economic advantages to a peasant family of large numbers of births.

They went on to comment that an agrarian low-income society "has a mortality and fertility pattern that fits pretty closely the conditions which Malthus thought . . . to be a universal tendency: high birth and high death rates. Growth of population is usually slow" (p. 10). This emphasis on the primary role of fertility decisionmaking was congenial in circumstances of twentieth-century interventions in such matters, but it does give rise to some anomalies. It implies that in traditional societies the only determinant of equilibrium in mortality–fertility levels was mortality, with fertility being raised to meet its levels. It ignores the more hopeful formulation of Malthus that marriage restraint by reducing fertility could in turn reduce the force of the positive checks and hence the mortality levels, resulting in a more tranquil society with moderate fertility and mortality levels. Coale and Hoover's is not strictly an equilibrium model because it allows for societies that fail to keep fertility sufficiently high; but these disappear, leaving only societies where mortality alone determines both mortality and fertility levels. This is the Malthusian model without the advantage of the preventive check.

In the same book, Coale and Hoover offer a theory for contemporary developing-world societies, one where a reduction of the birth rate, achieved at least partly by government intervention, slows the rate of population growth. The gains from this are not confined to raising nutritional levels but extend to allowing investment in broad economic development.

Ester Boserup published, in 1965, *The Conditions of Agricultural Growth*, often taken to be a theory that dispenses with the need for the principle of population. Indeed, the author herself took this view:

> The neo-Malthusian school has resuscitated the old idea that population growth must be regarded as a variable dependent mainly on agricultural output. I have reached the conclusion . . . that in many cases the output from a given area of land responds far more generously to an additional input of labour than assumed by neo-Malthusian authors. If this is true, the low rates of population growth found (until recently) in pre-industrial communities cannot be explained as the result of insufficient food supplies due to overpopulation, and we must leave more room for other factors in the explanation of demographic trends . . . —medical, biological, political etc.—which may help to explain why the rate of growth of population in primitive communities was what it was. (Boserup 1965: 14)

In fact, although Malthus regarded the potential for food production as setting the bounds for population growth, he held that in many societies famines are rare because the positive checks hold population numbers below the ceiling imposed by potential food production. If the checks are reduced, the extra labor will indeed result in greater food production until the agricultural limit is reached, famine results, and, because of starvation and heightened levels of epidemic disease, population numbers decline. Boserup argues that there is an alternative path, namely that the subsistence crises force human beings to innovate by turning to discoveries already known but underutilized, and so change the agricultural mode of production to allow greater production from a given unit of land provided that there are greater labor inputs. The argument that population growth has not been steady over the millennia but that there have been periods of relatively fast growth is undoubtedly true. But whether this observation does more than add to Malthusian theory is far less certain. The fact that population rises rapidly only when new modes of cultivation allow food supplies to increase unusually fast seems to support the concept of a Malthusian food-based population ceiling. Boserup cannot mean that every famine produces a change in the modes of production, or else all of human history, or at least its changes in agricultural systems and vast increases in population numbers, would have taken centuries rather than millennia. Nor can she mean that population growth has been determined solely by free-floating mortality checks, determined by nothing beyond themselves, for, in this case, the human race might again have compressed its whole history or have become extinct.

Many anthropologists have been much taken with the view that primitive food-collecting and hunting populations were not in a high-mortality state of misery as demanded by Malthusian theory, but had access to such ample food that they spent relatively little time meeting their nutritional needs. This derived partly from Boserup's (1965) thesis that the simpler modes of cultivation needed smaller labor inputs, partly from Richard Lee and Irven DeVore's (1968) *Man the Hunter*, and partly from Marshall Sahlins's (1974) *Stone Age Economics*, although much of the argument can be found in Alexander Carr-Saunders's 1922 work, *The Population Problem*. Sahlins went so far as to describe this situation as primitive affluence. It is unlikely that these descriptions of the mortality and state of welfare among premodern hunting and gathering cultures are the equal of Malthus's. We have criticized the anthropological data sources and conclusions elsewhere (Caldwell, Caldwell, and Caldwell 1987: 31–32) and have identified a failure to fully comprehend the unrepresentative situation of niche hunting and gathering groups in the modern world, and a selective use of the older reports, as helping to present an unsound, and far too optimistic picture. Primitive affluence theorists are hard put to explain long-term near-stationary population among hunting and gathering populations, a group to which Malthus devoted considerable attention.

Malthus was actually closer to Notestein with regard to fertility and to Boserup with regard to mortality than either seemed to realize, as is shown by this passage from the 1830 modification of the 1824 supplement to the *Encyclopaedia Britannica*:

> The frequency of wars and the dreadful devastations of mankind occasioned by them, united with the plagues, famines, and mortal epidemics . . . must

> have caused such a consumption of the human species that the exertion of the utmost power of increase must, in many cases, have been insufficient to supply it; and we see at once the source of those encouragements to marriage and efforts to increase population which . . . distinguished the legislation and general policy of ancient times. (Malthus 1960: 41)

Finally, it is pertinent to ask which political theories induced developing-world leaders to try to control population growth. I have been associated over the last 30 years with interrelated projects seeking information from those involved at the political or governmental levels with these decisions.[2] There is little doubt that in the densely settled agrarian lands of South Asia the dominant factor was the concept of a race between population and food, and of the need for higher nutritional levels. Among the survivors of the pre-1947 Indian Civil Service, which subsequently provided the upper echelons of the administrative services of both India and Pakistan, and after 1970 of Bangladesh, attitudes, shared with the now expatriated British members of the service, were usually quite consciously Malthusian.

The genesis of this essay springs from that discovery. President Ayub Khan, when interviewed in early 1969, explained the emphasis his administration in Pakistan placed on curbing fertility in terms of feeding the people. Subsequent research on the origins and success of the Bangladesh family planning program showed that officials first in East Pakistan, and then in Bangladesh, were convinced that the family planning program had to be pursued vigorously because of the visible density of the population. The 1974–75 Bangladesh famine strengthened these feelings, just as memories of the 1943–44 Bengal famine had been a factor in the 1951 Indian decision to opt for a family planning program. Chinese officials explain their present program more often than not in terms of needing the capacity to feed themselves. Elsewhere, in Southeast Asia, North Africa, and sub-Saharan Africa and in other less densely settled countries, the justifications for programs are undoubtedly more complex with a greater emphasis on broader economic development. But in the successor states of ancient agrarian civilizations, Malthusian beliefs and fears, while diminished, remain of significance and ready to reappear when crisis threatens.

Notes

Most of the work underlying this essay was undertaken with Pat Caldwell, and some of it with Bruce Caldwell, Barkat-e-Khuda, and Indrani Pieris. Assistance has also been provided by Wendy Cosford, Jeff Marck, and Rita Tanusoski. This essay was prepared for the conference organized by the Australian National Academies Forum and the National Library of Australia to mark the 200th Anniversary of the publication of Malthus's *First Essay*.

1 Where possible the quotations from Malthus have been taken from *The Works of Thomas Malthus*, ed. E. A. Wrigley and David

Souden, Pickering, London, 1986, although the series edited by Patricia James, Cambridge University Press, 1989, and Donald Winch's version of Malthus's second edition (1803), Cambridge University Press, 1992, have also been consulted. Malthus 1960 (Mentor Books, *Three Essays on Population*, New American Library, New York) has been used for the 1830 version of the 1824 supplement to the *Encyclopaedia Britannica*.

2 This research began as a 1968–69 Population Council project and later was a Ford Foundation–supported project in 1981–83. It is reported in Caldwell and Caldwell (1986). The East Pakistan/Bangladesh interviews carried out from 1969 and again intensively in 1995–96 are reported in Khuda et al. (in press).

References

Ambirajan, S. 1976. "Malthusian population theory and Indian famine policy in the nineteenth century," *Population Studies* 30, no. 1: 5–14.

Bagehot, Walter. 1869. *Physics and Politics, or The Thoughts on the Application of the Principles of "Natural Selection" and "Inheritance" to Political Society*. London: Kegan Paul, Trench, Trübner.

Bonar, James. 1924 [1885]. *Malthus and His Work*. London: Allen and Unwin.

Boserup, Ester. 1965. *The Conditions of Agricultural Growth: The Economics of Agrarian Change under Population Pressure*. Chicago: Aldine.

Breton, Yves. 1983. "Malthus and underdevelopment," in Dupâquier, Fauve-Chamoux, and Grebenik 1983, pp. 247–253.

Brown, Lester R. 1991. "The new world order," in Lester R. Brown et al., *The State of the World 1991: A Worldwatch Institute Report on Progress Toward a Sustainable Society*. New York: Norton, pp. 3–20.

Caldwell, J. C. 1966. "Africa," in *Family Planning and Population Programs: A Review of World Developments*, ed. B. Berelson et al. Chicago: University of Chicago Press, pp. 163–181.

———. 1993. "The Asian fertility revolution: Its implications for transition theories," in *The Revolution in Asian Fertility: Dimensions, Causes and Implications*, ed. R. Leete and I. Alam. Oxford: Clarendon Press, pp. 299–316.

Caldwell, John and Pat Caldwell. 1986. *Limiting Population Growth and the Ford Foundation Contribution*. London: Pinter.

Caldwell, John, Pat Caldwell, and Bruce Caldwell. 1987. "Anthropology and demography: The mutual reinforcement of speculation and research," *Current Anthropology* 28, no. 1: 25–43.

Campbell, R. H. and A. S. Skinner. 1976. "General introduction," in Smith 1976, pp. 1–60.

Carey, John. 1992. *The Intellectuals and the Masses: Pride and Prejudice among the Literary Intelligentsia, 1880–1939*. London: Faber and Faber.

Carr-Saunders, Alexander M. 1922. *The Population Problem: A Study in Human Evolution*. Oxford: Clarendon.

Coale, Ansley J. and Edgar M. Hoover. 1958. *Population Growth and Economic Development in Low-Income Countries: A Case Study of India's Prospects*. Princeton: Princeton University Press.

Cunningham, J. R. 1941. "Education," in O'Malley 1941, pp. 138–187.

Davis, Kingsley. 1946. "Human fertility in India," *American Journal of Sociology* 52: 243–254.

———. 1951. *The Population of India and Pakistan*. Princeton: Princeton University Press.

Dubois, J. A. 1906 [1816]. *Hindu Manners, Customs and Ceremonies*, trans. H. K. Beauchamp. Oxford: Clarendon.

Dupâquier, J., A. Fauve-Chamoux, and E. Grebenik (eds.). 1983. *Malthus Past and Present*. London: Academic Press.

Ehrlich, Paul R. 1968. *The Population Bomb*. New York: Ballantine.
Fryer, Peter. 1965. *The Birth Controllers*. London: Secker and Warburg.
Ghosh, D. 1946. *Pressure of Population and Economic Efficiency in India*. Bombay: Oxford University Press for Indian Council of World Affairs.
Glass, D. V. 1953. "Malthus and the limitation of population growth," in *Introduction to Malthus*, ed. D. V. Glass. London: Watts, pp. 25–54.
Godelier, Maurice. 1983. "Malthus and ethnography," in Dupâquier, Fauve-Chamoux, and Grebenik 1983, pp. 125–150.
Hardin, Garrett. 1968. "The tragedy of the commons," *Science* 162: 1243–1248.
Hollingsworth, T. H. 1983. "The influence of Malthus on British thought," in Dupâquier, Fauve-Chamoux, and Grebenik 1983, pp. 213–221.
Huxley, Julian. 1960. "World population," in Malthus 1960, pp. 63–81. New York: Mentor (first published in *Scientific American*, March 1956).
India Famine Inquiry Commission. 1945. *Final Report*. Madras: Government Press.
India Health Survey and Development Committee [Bhore Committee]. 1946. *Report of the Health Survey and Development Committee*. Delhi: Government of India Press.
Indian Census. 1870. *Report of the Census of the Punjab 1868*. Lahore.
———. 1875. *Report on the Census of British Burma 1872*. Rangoon.
———. 1877. *Report on the Census of Native Cochin 1875*. Madras.
———. 1878. *Report on the Census of Bengal 1876*. Calcutta.
———. 1882. *Report on the Census of the Northwest Province and Oudh 1881*. Allahabad.
———. 1883a. *Report on the Census of the Baroda Territories 1881*. Byculla.
———. 1883b. *Report on the Census of the Presidency of Madras 1881*. Madras.
———. 1884. *Report on the Mysore Census 1881*. Bangalore.
———. 1893. *General Report on the Census of India 1891*, by J. A. Baines. London: Eyre and Spottiswoode.
———. 1924. *Report on the Census of India, 1921*, by J. T. Marten. Calcutta.
———. 1933. *Report, Census of India 1931*, by J. H. Hutton. Delhi: Government of India.
———. 1953. *Report on the Census of India 1951*, by R. A. Gopalaswami. New Delhi.
Kane, Penny. 1995. *Victorian Families in Fact and Fiction*. London: Macmillan.
Keynes, John Maynard. 1951. *Essays in Biography*. London: Rupert Hart-Davis.
Khuda, Barkat-e-, John C. Caldwell, Bruce K. Caldwell, Indrani Pieris, Pat Caldwell, and Shameem Ahmed. In press. "The global fertility transition: New light from the Bangladesh experience," in *Comparative Perspectives on Fertility Transition in South Asia*, ed. IUSSP Committee on Fertility and Family Planning. Oxford: Clarendon Press.
Lee, Richard B. and Irven DeVore (eds.). 1968. *Man the Hunter*. Chicago: Aldine.
Malthus, Thomas R. 1960 [1830]. "A summary view of the principle of population," in *Three Essays on Population*. New York: Mentor, pp. 13–59 (first published 1830 as a slightly abridged version of the 1824 supplement to the *Encyclopaedia Britannica*).
———. 1986a. [1798]. *The Works of Thomas Robert Malthus*, vol. 1, *An Essay on the Principle of Population*, ed. E. A. Wrigley and David Souden. London: William Pickering.
———. 1986b [1826]. *The Works of Thomas Robert Malthus*, vol. 2, *An Essay on the Principle of Population*, 6th edition, ed. E. A. Wrigley and David Souden. London: William Pickering.
———. 1989. *An Essay on the Principle of Population*, 2 vols., ed. Patricia James. Cambridge: Cambridge University Press.
———. 1992. [1803]. *An Essay on the Principle of Population*, 2nd edition, selected and introduced by Donald Winch. Cambridge: Cambridge University Press.
Mamoria, C. B. 1961. *India's Population Problem*. Allahabad: Kitab Mahal.
Meadows, Donella H., Dennis L. Meadows, Jørgen Randers, and William W. Behrens. 1972. *The Limits to Growth: A Report for the Club of Rome's Project on the Predicament of Mankind*. London: Earth Island.
Mill, John Stuart. 1848. *Principles of Political Economy with Some of Their Applications to Social Philosophy*. London: Standard Library.

Mitchell, B. R. and Phyllis Deane. 1962. *Abstract of British Historical Statistics*. Cambridge: Cambridge University Press.

O'Malley, L. S. S. (ed.). 1941. *Modern India and the West: A Study of the Interactions of Their Civilizations*. London: Oxford University Press.

Ortega y Gasset, José. 1932. *The Revolt of the Masses*. New York: Norton.

Place, Francis. 1994 [1822]. *Illustrations and Proofs of the Principle of Population: Being the First Work on Population in the English Language Recommending Birth Control*. London: Routledge/Thoemmes.

Rawlinson, H. G. 1941. "Indian influence on the West," in O'Malley 1941, pp. 535–566.

Russell, E. John. 1954. *World Population and Food Supplies*. London: Allen and Unwin.

Sahlins, Marshall. 1974. *Stone Age Economics*. London: Tavistock.

Sanger, Margaret. 1938. *An Autobiography*. New York: Norton.

Sax, Karl. 1955. *Standing Room Only: The Challenge of Overpopulation*. Boston: Beacon.

Seeley, John. 1885. *The Expansion of England*. London.

Shah, K. T. (ed.). 1947. *Population: Report of the Population Sub-committee of the National Planning Committee*. Bombay: Vora.

Singh, Baljit. 1947. *Population and Food Planning in India*. Bombay: Hind Kitabs.

Smith, Adam. 1976 [1776]. *An Inquiry into the Nature and Causes of the Wealth of Nations*. Oxford: Clarendon.

Vogt, William. 1949. *Road to Survival*. London: Gollancz.

Wattal, P. K. 1916. *The Population Problem in India: A Census Study*. Bombay: Bennett, Coleman and Co.

———. 1958. *Population Problem in India: A Census Study*. New Delhi: Minerva.

Winch, Donald. 1992. "Introduction," in Malthus 1992, pp. vii–xxiii.

Wrigley, E. A. 1983. "Malthus's model of a pre-industrial economy," in Dupâquier, Fauve-Chamoux, and Grebenik 1983, pp. 111–124.